

Question 3: Singular Value Decomposition

(a) Find the Singular Value Decomposition of $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

Step 1: Compute $A^T A$

$$A^T A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Step 2: Find the eigenvalues of $A^T A$ We solve the characteristic equation $\det(A^T A - \lambda I) = 0$:

$$\begin{vmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 4 = \lambda^2 - 4\lambda = 0$$

The eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 0$.

Step 3: Determine Singular Values (Σ) The singular values are the square roots of the eigenvalues: $\sigma_1 = \sqrt{4} = 2$ and $\sigma_2 = 0$.

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

Step 4: Find the right singular vectors (V) For $\lambda_1 = 4$: $(A^T A - 4I)\mathbf{x} = 0 \implies \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
For $\lambda_2 = 0$: $(A^T A - 0I)\mathbf{x} = 0 \implies \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

Step 5: Find the left singular vectors (U) For $\sigma_1 = 2$: $\mathbf{u}_1 = \frac{1}{\sigma_1} A \mathbf{v}_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

For $\sigma_2 = 0$, we choose $\mathbf{u}_2$ such that it is orthogonal to $\mathbf{u}_1$: $\mathbf{u}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

$$U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Final SVD: $A = U \Sigma V^T$